

Coding & Solution

Date	30 June 2026
Team ID	SWTID-2026-9097
Project Name	Voice Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Kirti3005/Voice--Based--Concept--Understanding--Analyzer--SWTID-2026-9097
Programming Language(s)	Python
Framework(s) Used	Fast API, Streamlit, Hugging Face Transformers

Key Features Implemented	Voice-to-Text Transcription, AI-Based Concept Understanding Analysis, Personalized AI Feedback, Interactive Streamlit Dashboard, Performance Visualization & Database Integration
Pending / Incomplete Features	None
Setup / Run Instructions	<pre> pip install -r requirements.txt uvicorn main:app --reload streamlit run app.py </pre>



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes